



Beyond the screen: The effects of viewing environment and prior knowledge on learning in agent-based VR lecture videos

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Abstract

As an emerging video technology, 360-degree virtual reality (VR) video has garnered increasing attention due to its potential to create an immersive learning experience. However, the adoption of this immersive technology faces challenges related to the educational design and content production, and it remains unclear how the viewing environment of VR videos and learners' prior knowledge influence learning. This study employs a two-stage research design to address these issues. In Stage 1, an embodied agent-based VR lecture system is developed through an iterative design process to facilitate the effortless creation of VR lecture videos, and a case of a geography VR lecture video was generated to demonstrate its benefits. The usability evaluation supports the potential of the system for creating usable VR lecture videos. In Stage 2, a total of 60 participants with varying levels of prior knowledge viewed the generated VR lecture video using a head-mounted display (HMD, $n=30$) or a desktop screen (DS, $n=30$) to explore the effects on learning performance and perceptual experiences. Results showed that (a) learners with high prior knowledge reported better satisfaction, presence, and agent evaluation than those with low prior knowledge; (b) learning in VR lecture video via HMD has also enhanced satisfaction, presence, and agent evaluation than DS, and (c) these positive effects were not moderated by the learner's prior knowledge. The results indicate that the learning benefits of agent-based VR lecture videos brought by high-immersive viewing environments are applicable across learners with different prior knowledge levels. Overall, this study provides an educational design solution for VR lecture videos, and the findings underscore the value and general usability of this solution in assisting educators in developing immersive instructional media suitable for learners with diverse prior knowledge.

Keywords Lecture animation · Embodied agent · Virtual human · Multimedia learning · Virtual reality · 360-Degree panorama

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1 Introduction

In recent years, immersive virtual reality (VR) has been increasingly adopted in education settings. Previous literature has highlighted the benefits of appropriately designed VR content in education, including enhancing learners' performance, sense of presence, motivation, engagement, and satisfaction (Jongbloed et al., 2024; Klingenberg et al., 2020; Luo et al., 2021; Yin et al., 2024). One of the useful emerging VR technologies is the VR video based on 360-degree spherical panoramic (Evens et al. 2023; Li et al. 2023b; Rosendahl and Wagner 2024), which has been attracting increasing attention from researchers and educational practitioners, as this form of video media provides possibilities for creating immersive learning experiences, such as virtual field trips and immersive lectures (K.-H. Cheng & Tsai, 2019; Makransky & Mayer, 2022). Compared to the traditional 2D lecture videos, where information presentation is limited by the restricted field of view (FOV) or the selection of what is observable by the person holding the camera, 360-degree lecture videos offer promising innovations to overcome these viewing agency challenges brought by a fixed viewing perspective (Gold & Windscheid, 2020). It allows viewers to freely decide which parts of the image to focus on. Currently, 360-degree VR videos have been successfully used in various educational settings, including science education (Han, 2020), preservice teacher training (Gold & Windscheid, 2020; Kosko et al., 2022), geography instruction (Makransky & Mayer, 2022), and construction (Wen et al., 2022), as well as safety education (Araiza-Alba et al., 2021).

However, constrained by the limited accessibility and editing complexity of VR materials, one of the primary obstacles to adopting this type of VR content in education lies in the scarcity of content and essential technical skills, which poses a significant challenge for educators (Evens et al., 2023; Jensen & Konradsen, 2018). Specifically, the creation of 360-degree VR videos typically requires expensive camera equipment, leading to high production costs; meanwhile, when such 360-degree VR videos are used independently in common online learning scenarios, they also require additional embedded instructional guidance and explanations to meet the applicability of educational design (Evens et al., 2023). The low-cost creation and educational design of the VR video content remain obstacles to the widespread adoption of VR in an educational context, restricting the full realization of its educational advantages and value.

A common design convention in current lecture videos is to add embodied pedagogical agent (PA) elements as instructional components (Li et al., 2016), which provides inspiration for the educational design of 360-degree VR videos. These on-screen agents can be rendered as 2D/3D animations, real humans, or cartoon characters in learning environments (Davis et al., 2024; Davis & Vincent, 2019; Makransky et al., 2019b; Veletsianos, 2012). PAs can convey learning content and information through verbal and non-verbal social cues to provide cognitive and emotional support for learners (Lin et al., 2013; Schroeder et al., 2017), and it can enhance social presence, reduce feelings of isolation, and maintain high motivation in remote learning (Lawson & Mayer, 2022; Liew et al., 2020; Wang et al., 2022; Wang et al., 2023b). This approach is supported by Social Agency Theory (SAT; Mayer et al., 2003), which posits that embodied PAs in multimedia learning environments facilitate

learners' social presence and cognitive engagement, thereby achieving meaningful learning. However, compared to agent-enhanced 2D lecture videos, educators rarely integrated the PA into immersive VR media (Dai et al., 2022; Petersen et al., 2021), facing difficulties in creating PAs within the medium. The effectiveness of learning outcomes and experiences resulting from this embodied agent-enhanced VR video approach remains unclear.

Moreover, VR videos are typically viewed using two immersion types of delivery devices (Gold & Windscheid, 2020): desktop screen (DS) or head-mounted display (HMD), and the impact of the immersion principle of the viewing environment on VR-based learning has been widely investigated (e.g., Makransky & Mayer, 2022). However, the interaction between viewing environment and individual characteristics on VR learning has yet to be explored. As an important individual characteristic in the field of multimedia learning, learners' prior knowledge is a potential factor and important consideration in the design of instructional videos (Pi et al., 2023), since learners can utilize their prior knowledge to select, organize, and process relevant information in multimedia learning (Braune & Foshay, 1983). The knowledge-is-power hypothesis also posits that individual differences in domain-specific prior knowledge are among the strongest positive determinants of learning (Schneider & Simonsmeier, 2025). Meanwhile, the expertise reversal principle emphasizes the role of prior knowledge in learning from another perspective to explain this alternative (Kalyuga, 2014). It suggested design principles that are effective for novice learners may be ineffective or even hinder learning for more knowledgeable learners in many situations (Kalyuga, 2014). From this perspective, the different visual effects caused by the viewing environment of VR video might influence the learning process and outcomes of learners with different prior knowledge, and the interaction between these two factors influencing learning from VR lecture videos needs further exploration.

In this context, our research presents a new educational design approach that utilizes agent-based VR lecture videos to create immersive educational experiences. To support the adoption of this immersive educational media, we have developed a low-cost, user-friendly system that enables educators through an iterative design process to create immersive instructional content without requiring specific technical expertise. Most importantly, this study examined how the viewing environment of VR lecture videos and learners' prior knowledge affect learning, aiming to provide insights for designing immersive media and to promote the development of an immersive learning ecosystem that benefits diverse learners.

2 Literature review

2.1 360° video in educational settings

Lecture videos have long demonstrated significant value in promoting the development of distance education (Seo et al., 2021), and the online lecture videos have become a popular form of multimedia learning content (Kim et al., 2011). In recent years, with the advent of VR technology, immersive media content has begun

to play a role in education, holding the potential to facilitate diverse online learning experiences. Compared to traditional 2D media, VR media contribute to enhanced learning performance, sense of presence, and task engagement and induce empathy (Guan et al., 2024; Jongbloed et al., 2024; Luo et al., 2021). One widely applied form of immersive VR media is the 360-degree video, also known as spherical video or omnidirectional video (Li et al., 2023b), which represents a novel form of linear moving images (Gold & Windscheid, 2020). Typically, this VR video format constitutes an immersive environment built upon panoramic footage captured by omnidirectional cameras or rendered using 3D modeling software (Violante et al., 2019). A defining characteristic is that 360-degree video differs from 2D video with a fixed viewing perspective and allows viewers to freely determine the direction they watch within the video at any time (Gold & Windscheid, 2020). This capability for viewpoint manipulation presents unique technical advantages, liberating the camera operator from the need to predict which part of the image might be relevant, and the observer, instead of the videographer, decides what is important for the observation.

The advantages of 360-degree videos have been extensively discussed and validated across diverse educational contexts. For instance, Kosko et al. (2022) utilized 360-degree videos to investigate pre-service teachers' professional noticing of classroom interactions. Another study in safety education revealed that VR videos enhance learners' interest, enjoyment, and performance in topic-specific scenarios like coastal drowning prevention (Araiza-Alba et al., 2021). Despite these distinctive educational merits of VR video, educators often encounter challenges when designing and implementing VR panorama-based instructional scenarios. For example, the educational design and deployment of VR environments demands considerable time investment and technical expertise, coupled with a scarcity of teacher-friendly tools and pedagogical frameworks (Oubahssi et al., 2024). Furthermore, the high costs of recording equipment and content production, alongside the absence of embedded knowledge narration or instructional scaffolding, collectively hinder the full exploitation of 360-degree video's pedagogical advantages and potential. Several platforms or tools have emerged to simplify the creation and management of 360-degree VR content (e.g., Adobe Captivate VR, InstaVR, ThingLink, Krpano, and Marzipano), and their primary functionalities focus on enabling the integration of multimedia elements (e.g., embedding images, 2D videos, and interactive hotspots) into 360-degree images, as well as adding navigation links between different panoramic scenes. However, these platforms and tools often can't offer support for generating and processing 360-degree videos (except InstaVR), are not free (except Marzipano), and are not specifically designed for educational purposes, so they lack pedagogical agents intended to serve structured instructional objectives. Therefore, the design consideration is proposed based on the above:

Design consideration 1. More accessible tools are needed for the creation of educational VR videos to address the challenges related to educational design and content production.

2.2 Embodied pedagogical agent in the lecture video

To better apply VR videos for educational purposes, researchers have increasingly focused on their instructional design (Evens et al., 2023; Shadiev et al., 2022). Previous research has investigated the effects of redesigning VR video learning materials with visually enhanced features, such as visual signals or annotations (Albus et al., 2021), textual cues (Li et al., 2023a), or hotspots (Eiris et al., 2022), to facilitate learning. Recent years have seen researchers recognize the significance of PAs in learning environments (Wang et al., 2023b), making their integration into immersive VR media a critical design consideration. PAs—virtual characters in multimedia learning environments—are alternatively termed embodied agents, animated agents, virtual humans, and embodied or virtual instructors (Dai et al., 2022; Horowitz & Mayer, 2021; Li et al., 2024). It delivers learning support and affective experiences to learners through social cues across both visual and auditory channels (Ba et al., 2021), such as voice, expression, gestures, and gaze. Over the past few decades, PAs have been studied as a design approach to enhance lecture videos (Siegle et al., 2023), with research exploring their role in strengthening perceived social connections between computers and learners to optimize learning processes (Lin et al., 2013). Notably, computer-generated PAs can serve as virtual instructors to deliver presentations in lecture videos. This approach eliminates the need for live teacher recordings, allows precise control over teaching gestures and speech rate, reduces production costs, and enables the efficient creation of lecture videos that meet educational needs. For instance, Choi et al. (2024) developed an Animation System for Agent-based Presentations (ASAP), which synthesizes virtual agent-based lecture videos by mimicking the motions and speech of human instructors in lecture videos, with its efficacy validated through user studies involving both experts and novices.

A series of empirical studies has also revealed the positive impact of social cues such as anthropomorphized voice, facial expressions, and gestures presented by PAs in lecture videos on learners' emotional states and learning performance (Baylor & Kim, 2009; Craig et al., 2015; Davis et al., 2024; Kim et al., 2007; Kim & Baylor, 2006). For instance, agents displaying positive facial expressions during video-based learning significantly improve learners' knowledge acquisition (Wang et al. 2023a); Another research indicates that high-quality text-to-speech (TTS) systems can achieve comparable learning effectiveness and trustworthiness to human-voice recorded lecture videos (Chiou et al., 2020), and agents with a positive tone and visual appearance further provide better emotional engagement and instructional efficiency for learners in lecture videos (Ba et al., 2021). Additionally, when PAs establish eye contact with learners and employ pointing gestures, these behaviors positively correlate with enhanced knowledge retention and learning (Li et al., 2023c). The design consideration is proposed based on the above:

Design consideration 2. Enhancing lecture videos with the embodied PA represents a promising approach in the educational design of VR videos.

2.3 The combined potential of embodied pedagogical agent and VR lecture video

Several theoretical frameworks have been developed to explain and guide the design of embodied PAs, with predictive theories substantiating the potential benefits of incorporating such agents in VR lecture videos. For instance, Social Agency Theory (SAT; Mayer et al., 2003) posits that learners can be activated by embodied agents' social cues, thereby enhancing perceived social presence and cognitive engagement to facilitate meaningful learning. The Cognitive-Affective-Social Theory of Learning in Digital Environments (CASTLE; Schneider et al., 2022) similarly emphasizes the pedagogical value of social cues in technology-mediated learning. It suggests that digital social cues not only activate learners' social schemas (with activation intensity scaling with cue salience) but also significantly modulate ancillary cognitive processes during learning. Concurrently, Mayer (2014) proposed design principles for social cues in multimedia learning by reviewing the empirical studies of PAs, suggesting that personalization, voice, image, and embodiment cues can evoke learners' sense of social presence and stimulate learners' social responses, leading to deeper cognitive processing and better learning.

Furthermore, current research has incorporated PAs as design elements in VR learning environments, and the VR agent exhibits distinctive advantages in immersive learning environments. For instance, Grivokostopoulou et al. (2020) demonstrated that embedding a PA in virtual environments enhances both student experience and academic performance. Another study conducted by Wen et al. (2022) applied a PA as a guide and educational tools to facilitate students' learning of architecture-related knowledge during a 360-degree virtual field trip. Moreover, research shows that participants experienced a higher sense of presence when coexisting with virtual characters in fully immersive VR environments than in non-immersive viewing platforms (Guimarães et al., 2020).

In short, theories and empirical evidence substantiate the potential of incorporating the PA as an educational character that delivers information into 360-degree videos. However, there is limited research focusing on how to leverage PA to facilitate the creation of VR lecture videos and examine learners' learning outcomes and experiences with such videos, practices that could hold practical value for educators. The following exploratory hypothesis is proposed based on the above discussion:

Hypothesis 1. The higher-immersion viewing environment may lead to more favorable evaluations of the embodied PA than the lower-immersion condition.

2.4 Effects of the viewing environments and the learner's prior knowledge on VR-based learning

Viewing 360-degree videos requires specific devices equipped with compatible software. A common approach involves desktop computers or laptops equipped with specialized players for 360-degree videos, where viewing perspective and distance are typically controlled via mouse or keyboard, representing a non-immersive viewing mode. Another approach involves using head-mounted displays (HMDs), which

are associated with immersive VR experiences. For example, low-immersion solutions like Google Cardboard can transform smartphone displays into basic HMDs, utilizing cardboard as both viewing frame and device holder (Hebbel-Seeger, 2017); high-immersion solutions include dedicated HMDs from manufacturers like Meta Quest, Pico, and HTC VIVE. The use of HMDs could eliminate distractions from the external world when viewing VR content, creating the most immersive experience possible (Gold & Windscheid, 2020). Generally, compared to non-immersive desktop VR setups, immersive VR through HMDs enhances generative processing by delivering more authentic experiences, typically yielding stronger presence (Makransky et al., 2019a). Previous research has demonstrated numerous advantages of immersive media like VR panoramas in educational settings. For instance, higher levels of immersion provided by devices playing VR videos lead to better learning performance, interest, and presence (Rupp et al., 2019), and a high level of immersion is considered to significantly increase user satisfaction (Chang & Suh, 2025). However, some studies present divergent perspectives, suggesting VR media may induce higher cognitive load than desktop environments (Parong & Mayer, 2021), since immersive VR may distract learners and result in reduced learning outcomes and elevated cognitive load compared to PCs (Makransky et al., 2019a). Another study also found that VR elicited higher spatial presence but reduced recall than flat video and further emphasized that the learners' technology apprehension as an individual difference moderated the effect of learning media's modality features on spatial presence (Ahn et al., 2022). Thus, the effect of this enhanced immersion on learning continues to be actively debated in the literature; further investigation is warranted regarding how different viewing devices affect learning outcomes in 360-degree VR lecture videos. The following exploratory hypothesis is proposed based on the above:

Hypothesis 2. The higher-immersion viewing environment may lead to greater (a) learning satisfaction and (b) sense of presence and (c) learning performance, but may also result in higher (d) cognitive load than the lower-immersion condition.

Another critical factor requiring attention is the influence of learner characteristics on VR-based learning. An essential consideration in multimedia learning contexts is learners' prior knowledge, particularly in instructional video design (Pi et al., 2023). This is because design principles effective for novice learners may prove ineffective or even detrimental for more knowledgeable learners—a phenomenon known as the expertise reversal effect (Kalyuga, 2014). In multimedia learning, learners can utilize prior knowledge to select relevant information from visual cues and retrieve and integrate information from existing knowledge structures to construct mental models (Braune & Foshay, 1983). In this process, the individual's prior knowledge is a key factor in determining how visual representations affect learners' cognitive structures and learning processes, as prior knowledge can influence how easily learners perceive and interpret visual representations in working memory (Cook, 2006). From this perspective, the impact of different visual effects caused by VR viewing devices on learners' information processing may be moderated by prior knowledge, as low prior knowledge learners may struggle to correctly select, organize, and integrate information, while high prior knowledge allows learners to facilitate their learning

and experience in VR courses in meaningful ways (Makransky & Petersen, 2021). Meanwhile, the knowledge-is-power hypothesis also posits that individual differences in domain-specific prior knowledge are among the strongest positive determinants of learning (Schneider & Simonsmeier, 2025). Previous research has also shown the complex role of prior knowledge on learning, suggesting that learners' domain knowledge not only impacts the effectiveness of multimedia learning but also influences cognitive load, learning satisfaction, and emotional experience (Chen et al., 2014; Cheng et al., 2023; Shangguan et al., 2020). However, it is important to note that the effect of prior knowledge on learning outcomes is often complex and context-dependent. Previous research has indicated that this relationship is not always constant, as it can vary depending on the influence of numerous moderating variables (Simonsmeier et al., 2022). For instance, a study conducted in VR laboratory settings has found that learners with low prior knowledge benefit more from signaling used as visual cues, as signaling helps guide the attention of learners with low prior knowledge, effectively aiding them in selecting relevant information and reducing their cognitive load, while signaling was found to have no significant impact on the cognitive load, intrinsic motivation, or learning performance of learners with high prior knowledge (Han et al., 2023). Therefore, this study exploratively investigates the impact of learners' prior knowledge levels as an individual characteristic on VR lecture video learning conducted through different viewing devices. Understanding these individual differences' impact on VR lecture video learning can help instructional designers develop appropriate educational resources tailored to learners. In light of the aforementioned theoretical tensions, the following exploratory hypotheses are proposed based on the above:

Hypothesis 3. Learners with higher prior knowledge may exhibit (a) better learning performance and (b) learning satisfaction, as well as (c) lower cognitive load than those with lower prior knowledge.

Hypothesis 4. Prior knowledge may moderate the effects of viewing environment on (a) learning performance, (b) learning satisfaction, and (c) cognitive load.

2.5 Study outline

In summary, the aforementioned theories and research have demonstrated the potential of using immersive VR media and embodied agents to enhance learning content, yet educators face challenges in creating such immersive educational materials, and it remains unclear how the viewing environment of such VR media and learners' prior knowledge influence learning. Therefore, we conducted a two-phase study. Figure 1 illustrates the motivation behind this research and the relationship between its two sub-studies. Specifically, we initiated a design project in stage 1 based on design considerations 1–2 for an agent-based VR lecture video generation system, aiming to investigate educational design approaches for VR videos by integrating the PA as a virtual instructor, thereby enhancing educational benefits. Meanwhile, this system enabled the generation of VR lecture videos, thereby providing experimental materials for both the usability evaluation of the agent-based VR lecture video generation system and the subsequent empirical testing. Subsequently, building

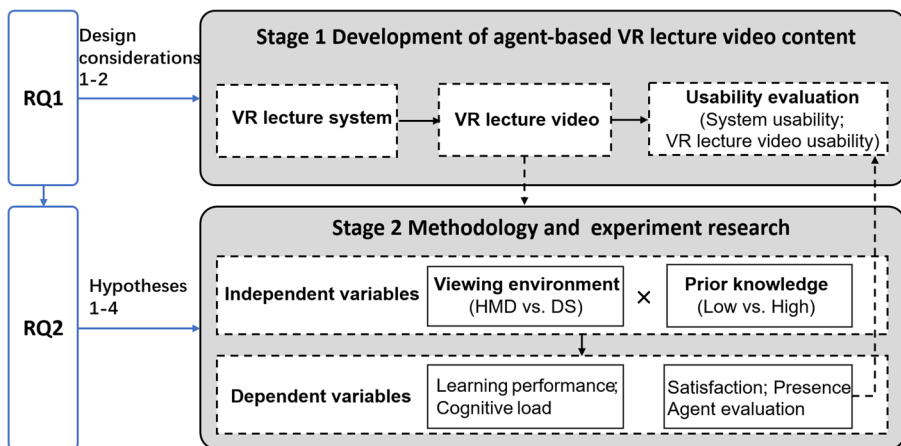


Fig. 1 Research overview

upon the outcomes of stage 1, stage 2 further investigated the impacts of viewing environments and prior knowledge on learning from agent-based VR lecture videos to test Hypotheses 1–4. These hypotheses were proposed to examine theoretically plausible yet competing perspectives within the VR learning literature. This two-phase research design establishes a direct link between technological design and instructional effectiveness, allowing us to examine both the universal applicability of this VR instructional medium and potential configurations matching specific viewing devices with learner profiles. Overall, our research follows a two-phase roadmap as illustrated in Fig. 1, and both phases are guided by the following research questions:

RQ1: How to design and develop a usable agent-based VR lecture tool to support the educational adoption of VR video?

RQ2: How do viewing environments (DS vs. HMD) and learners' prior knowledge levels (low vs. high) affect learning performance and perceptual experiences (including satisfaction, presence, agent evaluation, and cognitive load) with VR video?

3 System design overview

3.1 Approach

We conducted a convenient iterative design practice based on the design considerations 1 and 2, drawing inspiration from the design-based research process, informed by oral feedback from 4 subject-matter experts and 6 user evaluators to adjust and improve our project throughout the software development process. One cycle of the design process consists of the following six iterative phases (Easterday et al., 2014): (1) focus on the problem; (2) understand the problem; (3) define design goals; (4) conceive the outline of the solution; (5) build the solution; and (6) test the solution.

Overall, we followed these design steps sequentially, although certain phases were conducted in parallel (see Table 1). Specifically, the major difference between the design circles 1 and 2 lies in the redefinition of project goals from “The learning environment provides agent-enhanced 360-degree instruction” to “The editing tool generates agent-enhanced 360-degree lecture video.” This transformation is driven by the realization inspired by the 2 experts in design circle 1’s test phase in this study that learning through 360-degree videos requires highly structured knowledge representation, necessitating instructional scripts to organize relevant learning resources—something lecture videos are particularly well-suited for. Therefore, we opted to organize instructional resources through the way of micro-lectures rather than simply adding a PA into a 360-degree video to better support learning.

In short, this project aims to develop a VR lecture tool using the Unity engine to generate agent-based VR lectures. The system supports the local integration of multimodal learning resources—such as PowerPoint presentations, instructional scripts, and 3D animations—as well as the customization of embodiment cues for the pedagogical agent. It can automatically generate lecture videos in either 360-degree spherical VR format or traditional 2D media. The design of the PA follows Mayer’s embodiment principle (Mayer, 2014), which posits that learners achieve deeper

Table 1 Design process

Design stage	Design circle	
	Circle 1 iterative design phases	Circle 2 iterative design phases
Focus	Develop a service tool for 360-degree panoramic learning environments	Develop an editing tool for generating 360-degree panoramic instructional videos
Understand	Integrate the PA into 360-degree panoramic environments to deliver instructional content through verbal and non-verbal social cues	Drawing on the instructional scripting process of 2D lectures to develop an agent-enhanced instructional approach for VR lecture videos
Define	The learning environment provides agent-enhanced 360-degree VR instruction	The editing tool generates agent-enhanced 360-degree lecture video
Conceive & build	Render 360-degree videos on virtual spherical displays and add a head-tracked PA within the user’s FOV to provide instructional content	Supports the settings of custom PPT uploads, virtual scene transitions, slides playback with synchronized narration, and embodied agent control based on instructional scripts
Test	Conduct oral evaluation via focus groups with 2 subject-matter experts to derive iterative improvements	Conduct oral evaluation via focus groups with 2 subject-matter experts and 6 users to derive iterative improvements

understanding when on-screen agents exhibit humanlike gestures, movements, eye contact, and facial expressions (Cohen's $d=0.36$).

3.2 System architecture

Figure 2 shows the architecture of the VR lecture system development during design cycle 2. This system provides a graphical user interface (GUI) as an editor tool to support the customization of embodied agents, the alignment of multimodal learning resources based on instructional scripts, and the recording of 360-degree panoramic video. The technical details for the major components are contained as follows.

- (1) **Scene & Asset.** The virtual scene involved in this project includes 3D classrooms and 360-degree panoramas. The plugin called ProBuilder was used to streamline the 3D classroom modeling process and maintain high-quality results. Meanwhile, this project adopts a panoramic sphere method for building the 360-degree panoramic scene asset. A total of 9 photographs captured from different spatial directions were stitched together to create a single panoramic image with a 2:1 aspect ratio. This 2D panoramic image was then used as a texture map for material creation, which was applied to the corresponding sphere object, completing the visualization of the spherical panorama along with annotations of geographical knowledge for education.
- (2) **Embodied agent.** This component focuses on the generation of PA's image appearance and control of PA's voice and embodiment cues. In PA's appearance module, this project integrates an avatar software development kit (SDK) of Ready Player Me into the lecture system. It supports users to create avatars based on a generic avatar model library or a single picture and also allows personalized customization for gender, skin, hair, glasses, clothing, expressions, and other appearances to meet the requirements for agent-based presentation in this lecture system. In PA's voice module, a TTS plugin called RTVoice was used to generate humanlike audio in real-time based on an instructional speech script corresponding to each slide, and the BlendShape animation was applied to smoothly adjust the PA's lip movements according to the voice. In PA's motion module, this project employed natural eye gaze, lecturing gestures, and body swaying as background actions

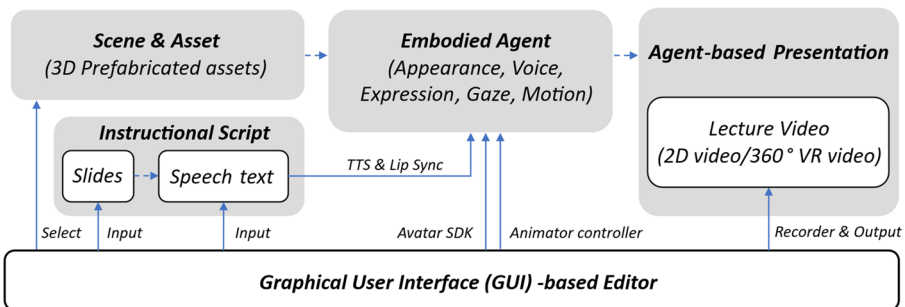


Fig. 2 System architecture

during speech to enhance the realism of the embodied agent, and an animation state machine is employed in conjunction with 3D limb bone movement animations for PA's action behaviors. Specifically, the gaze behavior was implemented by orienting the agent's head and eyes toward the viewer's location, as the prior study has demonstrated that an instructor's direct gaze is a design choice capable of simultaneously enhancing learning outcomes ($g=0.30$) and increasing parasocial interaction ($g=0.34$) in lecture videos (Kuang et al., 2023). Meanwhile, all body movements—including lecturing gestures and body sways—were implemented using pre-recorded humanoid animations combined with an animation state machine. In addition to default idle movements, we provided three optional gesture types: a greeting or farewell gesture, a general lecturing gesture, and a directive gesture. Users can select the appropriate gesture type to be triggered during the presentation of that slide when setting the lecture script corresponding to each slide. This approach aligns with our design goal of enabling educators to create VR lecture videos without requiring expertise in 3D animation.

- (3) Instructional script and agent-based presentation. This component focuses on script-based video generation for VR lectures. In terms of learning resource types, this project supports the unified local integration of multiple types of lecture elements, including PowerPoint slides, lecture speech text, and 3D simulation animation. Users can select local relevant files using a GUI editor. In terms of the VR lecture script, this project uses a slide-based presentation to align the slide pages with their corresponding speech text script. This method of aligning lecture elements is more efficient and better suited for creating educational presentations compared to the timeline method, as generally adopted in other animation or video production systems. In terms of VR lecture recording, the generation of video formats relies on Unity Recorder, which supports the rendering of both 2D flat videos and 360-degree panoramic videos in an 8 K resolution with a 2:1 aspect ratio, suitable for playback on VR headsets.

3.3 Case of VR lecture

After the design circle 2 of iterative design, we developed a variety of instructional resources related to contour lines and used the VR lecture system to integrate these resources based on the instructional script to create a VR lecture video (see Fig. 3). The reason for choosing this course is that the geography knowledge about contour lines is relatively abstract, and understanding these structures requires learners to combine various spatial, logical, and cognitive skills (Šikl et al., 2024), while VR can effectively concretize learning materials and convey them to students through 3D animations and virtual field trips. Moreover, to ensure optimal visual fidelity of the instructional text, the media content, and the embodied agent, the system was configured to output 360-degree videos at 8 K resolution. This design choice was made to address the pixel stretching encountered in panoramic videos, as converting standard flat video resolution to panoramic video requires mapping a flat image onto a spherical surface. Given that the viewer's perspective is positioned at the center of this spherical VR environment, the FOV they observe captures only a portion of the full spherical image. Consequently, 8 K resolution is typically required to ensure the nec-



(a) A frame screenshot of the VR lecture video



(b) The learner's field of view (FOV) at specific frames.

Fig. 3 VR lecture of contour line

essary pixel density to make an adequate clarity in VR panoramic videos. Figure 4 displays the instructional script. More script details can be found in Appendix A.

4 Methodology

4.1 Participants

We recruited 60 participants from a college in eastern China as evaluators, ranging from freshmen to sophomores ($M_{age}=18.83$, $SD=0.827$). Participants with normal hearing, visual acuity, or corrected vision were included in the subsequent evaluation activities. We obtained written informed consent from all participants before the experiment began to ensure they fully understood the research objectives, expected contributions, and the rewards for participation. The research protocol was approved

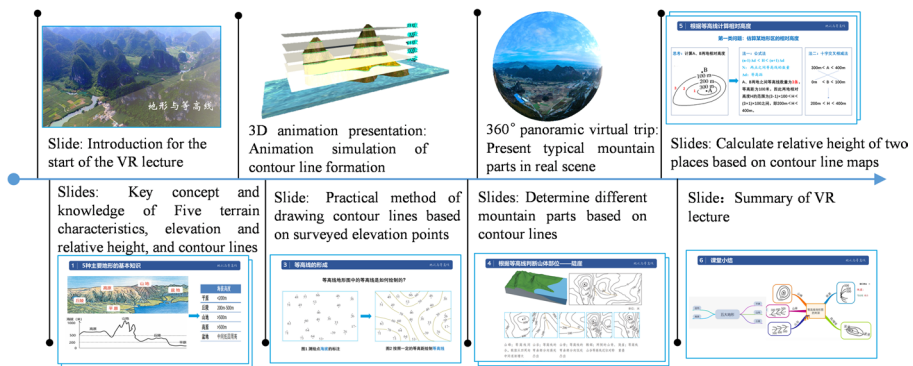


Fig. 4 Instructional scripts of VR lecture video

Table 2 Demographic profile of respondents

Items	Categories	Frequency (<i>N</i> =60)	Percentage (%)
Gender	Female	48	80%
	Male	12	20%
Age	18 years	26	43.3%
	19 years	18	30%
	20 years	16	26.7%
	21 years	0	0%
Familiarity with VR	Never use	16	26.7%
	Rarely use	35	58.3%
	Sometimes use	7	11.7%
	Often use	2	3.3%
	Very often use	0	0%

by the Institutional Review Board (IRB) of the university affiliated with the first author, and ethical considerations were integrated throughout the design, implementation, and reporting of the study. Demographic information of the participants is presented in Table 2.

Furthermore, in order to evaluate the agent-based VR lecture video generation system developed in stage 1, we additionally recruited 5 educators with experience in lecture video design and production to evaluate the usability of the system.

4.2 Research design

In this study, the independent variable was the viewing device (DS vs. HMD). Specifically, in the experimental group (HMD group, *N*=30), students used the HMDs to learn from the VR lecture video, while in the control group (DS group, *N*=30), students used the desktop computer. The dependent variables were learning performance and perceived experiences, including satisfaction, sense of presence, cognitive load, and agent evaluation, while the moderator variable was learners' prior knowledge levels (low vs. high). A 2×2 experimental factorial design was used, in which media environment type was crossed with the learners' prior knowledge.

4.3 Apparatus

The material used in this experiment is the 8 K micro-VR lecture video (duration: 10 min 16 s) created in the first research stage. For DS conditions, a computer lab equipped with PotPlayer, software capable of playing 360-degree VR videos without the VR headset, was used to enable participants to view the lecture videos on the computer screen. For HMD conditions, PICO Neo 3 headsets with controllers were employed, and these VR devices were equipped with a 120 Hz refresh rate. And it is user-friendly for nearsighted users with corrected vision.

4.4 Measurements

Two researchers familiar with the content developed the pre-test and post-test under the guidance of experts in the field. The pre-test instruments used in this study included a prior knowledge test and demographic survey, which collected information on participants' gender, age, academic year, major, prior experience with VR, and familiarity with VR, while the post-test instruments consisted of a learning performance test and a learning experience survey (including learning satisfaction, presence, cognitive load, and agent evaluation; see Appendix B).

The prior knowledge test and the learning performance test were developed based on the key concepts and knowledge points covered in the VR lecture video to ensure alignment between the test items and the instructional objectives, as well as to establish content validity. Additionally, the wording of all items was reviewed by both researchers and the curriculum expert to eliminate ambiguity. The prior knowledge test consisted of 9 multiple-choice questions and 7 true/false questions, all based on the topic of contour lines, with a total possible score of 16 points. The learning performance test comprised 25 questions, including 12 multiple-choice questions worth 1 point each, 5 multiple-choice questions worth 2 points each, and 8 true/false questions worth 1 point each, resulting in a maximum total score of 30 points. Cronbach's alpha values indicate that the prior knowledge test ($\alpha=0.70$) and the learning performance test ($\alpha=0.77$) have adequate internal consistency.

In terms of the cognitive load, we employed a two-item cognitive load scale developed by Paas and Van Merriënboer (1993) and recently revised by Ma et al. (2025), which was designed to provide an overall assessment of cognitive load, encompassing learners' mental load (i.e., How difficult do you think it was to learn in this VR lecture video?) and mental effort (i.e., How much effort do you think you devoted to the learning process in this VR lecture video?) during the learning process. These two items were rated by the participants on a 5-point scale ranging from 1 (very low cognitive load) to 5 (very high cognitive load), and this scale has been widely adopted in the field of multimedia learning (Sweller et al., 2019). The scale's internal consistency was assessed with Cronbach's $\alpha=0.91$.

Three constructs (i.e., presence, agent evaluation, and learning satisfaction) in the learning experience survey were used to measure the usability of the VR lecture video, corresponding to the subjective experience evaluations of the virtual scene, the embodied agent, and the overall quality of the lecture video. The presence in the virtual scene was measured using five items adapted from a previous study

(Klingenberg et al., 2020; e.g., While I was in the virtual environment, I had a sense of “being there”). Cronbach’s alpha for this 5-point Likert test was 0.93. The agent evaluation questionnaire was adapted after adding the item to measure the anthropomorphism of the embodied agent (Dai et al., 2024; e.g., Did the virtual instructor’s face move well while talking?), and each of these four items was administered on a 7-point Likert scale. Cronbach’s alpha was 0.95. The learning satisfaction of the VR lecture video was assessed using four items (Yin et al., 2024; e.g., was satisfied with this type of VR learning experience). Cronbach’s alpha for this 5-point Likert test was 0.92.

Moreover, we employed the system usability scale (SUS; Brooke, 1996) to evaluate the usability of the agent-based VR lecture video generation system (see Appendix B). The SUS consists of ten 5-point Likert items that capture key aspects of usability, such as ease of use, usefulness, perceived complexity, consistency, and learnability, and it enables the measurement of a system’s perceived usability with a relatively small sample of users (Brooke, 2013). The scale demonstrated a Cronbach’s alpha of 0.60.

4.5 Procedure

The entire process was conducted in a VR classroom setting and lasted approximately 60 min, which included four main stages (as shown in Fig. 5): (a) Prepare, (b) a brief introduction, (c) watch the VR lecture video, and (d) complete the evaluation questionnaire. During the experiment, two researchers supervised the course and data collection procedure. First, the researcher provided participants with instructions on procedures and operational training, and then randomly assigned each participant to one of the two treatment conditions. After providing informed consent, participants were asked to independently complete an untimed pre-test on prior knowledge. Next, the researchers guided each participant to watch the VR lecture video corresponding to their assigned experimental condition. Finally, participants independently completed, in sequence, the learning experience survey (including satisfaction, presence, cognitive load, and agent evaluation) and the learning performance test, both of which were untimed. Participants received financial compensation for their participation. Additionally, five educators completed the system usability scale after being systematically familiarized with the functionalities of the agent-based VR lecture video generation system.

In the aforementioned experimental procedure, we implemented several strategies to mitigate both the risk and impact of cybersickness: (a) The VR lecture video adopted a stable camera perspective to avoid rapid motion, as joystick-based movement or teleportation often induces more cybersickness (Eudave & Martínez, 2025); (b) Participants remained seated throughout the experiment and were able to view the 360-degree VR videos by making natural adjustments to their body and head orientation; (c) The total duration of VR exposure was kept short (approximately 10 min); (d) Participants were informed of their right to withdraw from the experiment at any time. In this study, no participants reported discomfort, nausea, or dizziness, and all completed the experimental procedure without interruption.

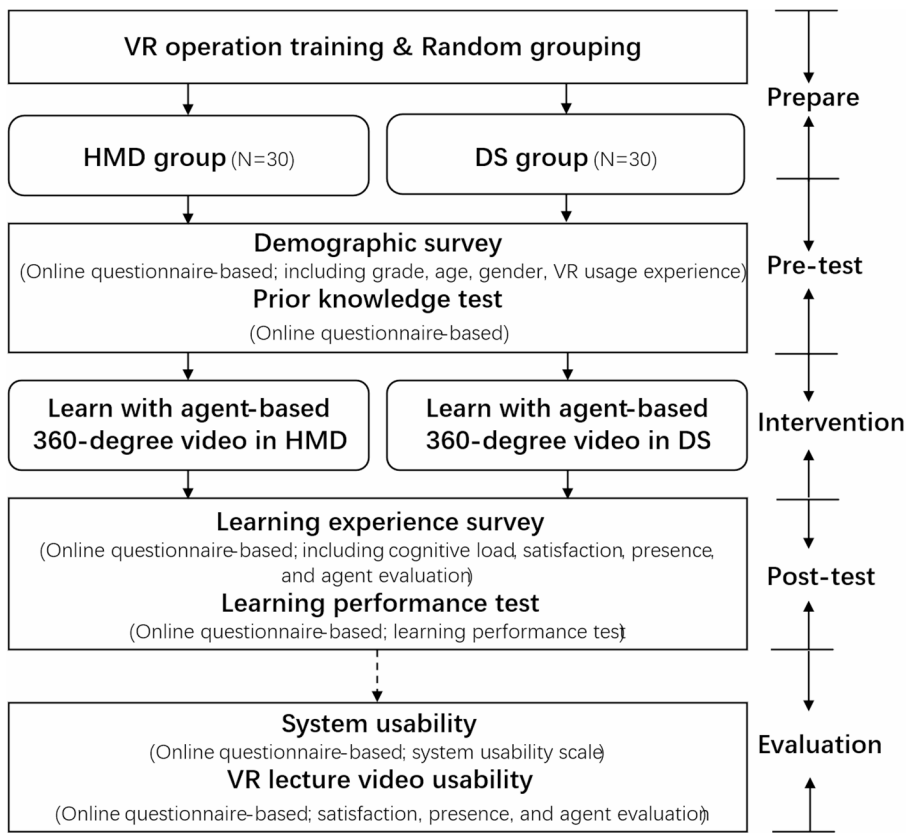


Fig. 5 Experimental process in this study

5 Results

The χ^2 tests and independent-samples t-tests were employed to assess the equivalence of demographic variables across conditions. The χ^2 test indicated no significant differences in gender distribution across groups ($\chi^2 (N=60) = 0.417, p = 0.519$). The t-test results showed no significant differences between the experimental group (group A) and control group (group B) in terms of age ($t (58) = 0.936, p = 0.353$), VR usage experience ($t (58) = 0.895, p = 0.374$), or prior knowledge ($t (58) = 0.579, p = 0.565$). It indicated that the groups were similar in their basic characteristics.

5.1 Usability evaluation

5.1.1 System usability

Table 3 presents the usability scores of the system. The SUS scores were converted into a range of 0 to 100 points based on the usage program set by Brooke (1996). Overall, the system achieved a mean SUS score of 76.67. According to the grading

Table 3 Means and standard deviations of the system usability

Evaluator	Rating										Total score
	1	2 (R)	3	4 (R)	5	6 (R)	7	8 (R)	9	10 (R)	
1	4	2	4	3	4	2	4	2	4	2	72.5
2	3	1	5	2	4	1	5	1	4	1	87.5
3	4	2	4	2	5	3	4	3	2	3	65
4	4	2	5	2	5	2	5	2	5	2	85
5	3	1	3	1	3	2	3	1	4	1	75

criteria proposed by Bangor et al. (2008), this score falls within the “good” range, indicating that educators generally perceived the system as having good usability. Among the items, the highest mean scores ($M=4.0$) were observed for items 3, 5, and 7, indicating that the system was perceived as easy to use, functionally integrated, and quick to learn. In contrast, items 4 and 6 received the lowest mean scores ($M=3.0$) after being reverse-coded, suggesting that the system’s consistency and technical support require further improvement.

5.1.2 VR lecture video usability

Table 4 presents the means and standard deviations of 60 participants’ scores under the HMD and PC conditions. As shown in Table 4, participants provided positive evaluations for the VR lecture video, and the learning performance, learning satisfaction, sense of presence, and agent evaluation achieved the highest scores in the HMD condition, whereas cognitive load was highest in the PC condition. Specifically, participants reported above-average presence ($M=17.60$, $SD=5.11$ on a 25-point cumulative scale) after viewing the virtual scenes of the VR lecture video, and this immersive educational media elicited strong presence ($M=20.50$, $SD=2.95$) when presented in the HMD condition. Participants’ evaluation of the embodied agent was acceptable ($M=20.00$, $SD=5.71$). As shown in Fig. 6, among the four sub-items assessing agent characteristics, although all mean ratings across participants exceeded 4 (out of 7), only perceived friendliness scored above 5 (out of 7); however, when viewing in the HMD condition, all agent characteristic ratings exceeded 5 (out of 7). The results of agent evaluation indicate that, although our agent is not an interactive intelligent agent, its presence—rendered within the media environment of the VR lecture video—combined with general body movements, lecturing gestures, and voice–lip synchronization, achieved acceptable evaluation outcomes. Moreover, this agent-based VR lecture video was also received favorably, as participants reported

Table 4 Means and standard deviations of the dependent variables

Measurements	HMD ($N=30$)		DS ($N=30$)		Total ($N=60$)	
	M	SD	M	SD	M	SD
learning performance	21.67	5.03	19.57	4.73	20.62	4.96
Learning satisfaction	18.10	2.45	14.10	2.91	16.10	3.34
Presence	20.50	2.95	14.70	5.20	17.60	5.11
Cognitive load	4.70	1.62	5.13	1.98	4.92	1.81
Agent evaluation	22.57	4.67	17.43	5.56	20.00	5.71

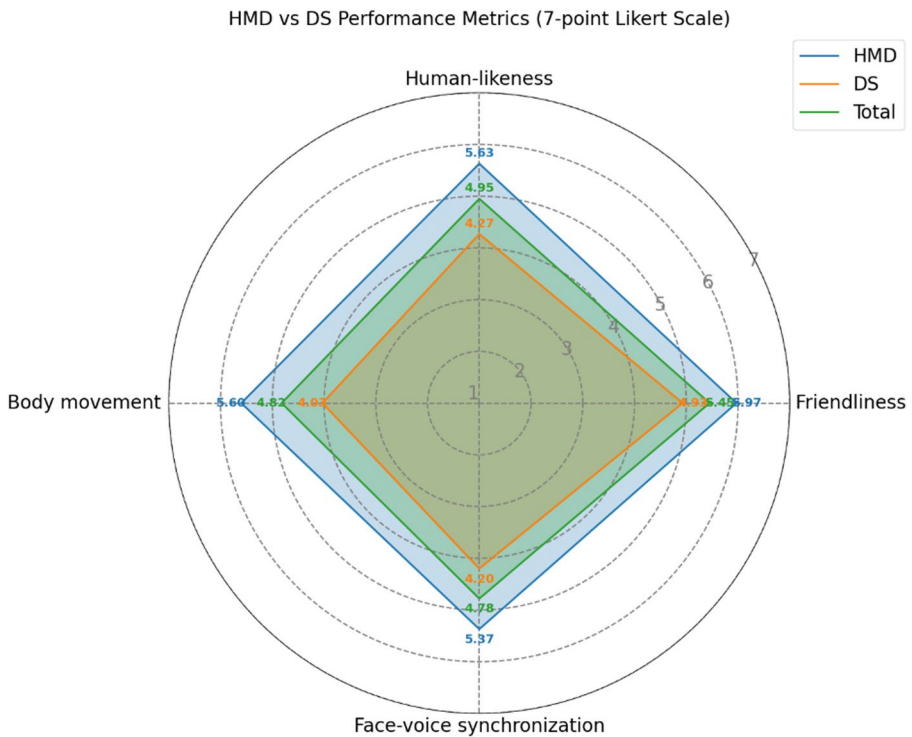


Fig. 6 Comparison of agent characteristics for the different viewing environment

high learning satisfaction ($M=16.10$, $SD=3.34$ on a 20-point cumulative scale). Overall, these results preliminarily indicate the usability and potential of the system developed in this study for creating VR lecture videos.

5.2 Effects of viewing environment and learners' prior knowledge on measurement outcomes

To further examine the effects of different viewing environments (HMD vs. DS) and to explore whether learners' prior knowledge levels influence learning and perceived experiences under different viewing conditions, we divided participants into a high prior knowledge (HPK) group ($n=32$) and a low prior knowledge (LPK) group ($n=28$) based on the median score of the prior knowledge test. A two-way ANOVA was then conducted exploratorily to examine the potential influence of prior knowledge level in this study (as shown in Table 5). We used η_p^2 as the measure of effect size (large: $\eta_p^2 \geq 0.14$; medium: $0.14 > \eta_p^2 \geq 0.06$; small: $0.06 > \eta_p^2 \geq 0.01$; Cohen, 1988).

5.2.1 Learning performance

As shown in Table 4, the mean score on the learning performance test was 20.62 ($\text{Min}=7$, $\text{Max}=30$, $SD=4.96$), with a correct response rate of 68.7%, indicating a

Table 5 Means and standard deviations on dependent variables of learning performance and experience survey for four conditions

Measurements	Descriptive analysis M (SD)				Significance		
	HMD		DS		Envi- ronment type	Prior knowledge	Inter- action
	HPK	LPK	HPK	LPK	<i>p</i> -value	<i>p</i> -value	<i>p</i> -value
learning performance	21.17 (5.40)	22.42 (4.54)	20.29 (4.30)	18.94 (5.13)	0.096	0.970	0.317
Learning satisfaction	18.56 (2.26)	17.42 (2.68)	15.00 (2.83)	13.31 (2.82)	<0.001	0.045	0.692
Presence	21.11 (2.81)	19.58 (3.03)	16.57 (4.07)	13.06 (5.64)	<0.001	0.021	0.355
Cognitive load	4.67 (1.91)	4.75 (1.14)	5.93 (1.94)	4.44 (1.79)	0.306	0.131	0.092
Agent evaluation	23.67 (4.46)	20.92 (4.68)	19.43 (4.26)	15.69 (6.10)	0.001	0.015	0.703

Bold values are significant at $\alpha = 0.05$

moderate level of difficulty. The overall score distribution exhibited a skewness of -0.524 and a kurtosis of -0.239, and the Shapiro–Wilk test yielded $p=0.055>0.05$, suggesting that the learning performance scores were normally distributed. Specifically, scores above 25 accounted for 23.3% of the sample, while scores below 15 accounted for 13.3%, indicating that the test was not affected by ceiling or floor effects. As shown in Table 6, the two-way ANOVA indicated that there were not significant main effects of environment type ($F(1,56)=2.871$, $p=0.096$, $\eta_p^2=0.049$) or prior knowledge level ($F(1,56)=0.001$, $p=0.970$, $\eta_p^2<0.001$) indicating that Hypotheses 2c and 3a were not supported. These results suggest that, in this study, neither the viewing environment nor students' prior knowledge significantly influenced learning performance from the VR lecture video, indicating the adaptability and broad applicability of this educational VR video format across different conditions. Meanwhile, the result anticipated by Hypothesis 4a was not observed, as there was no significant interaction effect between the two factors ($F(1,56)=1.019$, $p=0.317$, $\eta_p^2=0.018$). This indicates that the VR lecture video used in this study exerts a non-differential influence on learners' performance regardless of viewing device and is equally effective for learners with varying levels of prior knowledge.

5.2.2 Learning satisfaction

Consistent with Hypothesis 2a, the results indicated a significant main effect of environment type on learning satisfaction with the VR lecture video ($F(1,56)=30.975$, $p<0.001$, $\eta_p^2=0.356$). Additionally, consistent with Hypothesis 3b, there was a significant main effect of prior knowledge level ($F(1,56)=4.217$, $p=0.045$, $\eta_p^2=0.070$). However, there was no significant interaction effect between environment type and prior knowledge level ($F(1,56)=0.159$, $p=0.692$, $\eta_p^2=0.003$), which is inconsistent with the prediction of Hypothesis 4b. These results suggest that watching the VR lecture video via HMD leads to higher learning satisfaction, and learners with

higher prior knowledge tend to report greater satisfaction compared to those with lower prior knowledge.

5.2.3 Presence

Consistent with Hypothesis 2b, the results indicated a significant main effect of immersion type on presence ($F(1, 56)=27.087, p<0.001, \eta_p^2=0.326$). Notably, an unexpected finding was that the main effect of prior knowledge level was also significant ($F(1, 56)=5.617, p=0.021, \eta_p^2=0.091$). Additionally, there was no significant interaction effect between the two factors ($F(1, 56)=0.869, p=0.355, \eta_p^2=0.015$). These results indicate that watching the VR lecture video via HMD significantly enhances learners' sense of presence during the learning process. Moreover, learners with higher prior knowledge reported higher levels of presence compared to those with lower prior knowledge.

5.2.4 Cognitive load

The main effects of immersion type ($F(1, 56)=1.069, p=0.306, \eta_p^2=0.019$) and prior knowledge level ($F(1, 56)=2.350, p=0.131, \eta_p^2=0.040$) on cognitive load were not significant, nor was their interaction effect ($F(1, 56)=2.939, p=0.092, \eta_p^2=0.050$), indicating that the hypothesized relationships in Hypotheses 2d, 3c, and 4c were not supported. These results suggest that when the VR lecture video used in this study is employed as a learning material, neither the viewing device nor learners' prior knowledge significantly influences cognitive load.

5.2.5 Agent evaluation

Consistent with Hypothesis 1, the results indicated a significant main effect of environment type ($F(1, 56)=13.412, p=0.001, \eta_p^2=0.193$). An unanticipated finding was that prior knowledge level also had a significant main effect on agent evaluation ($F(1, 56)=6.305, p=0.015, \eta_p^2=0.101$). However, there was no significant interaction effect between environment type and prior knowledge level ($F(1, 56)=0.147, p=0.703, \eta_p^2=0.003$). These results suggest that participants who viewed the VR lecture video via HMD rated the PA more positively than those using DS. Furthermore, learners with higher prior knowledge tended to provide higher evaluations of the agent compared to those with lower prior knowledge.

6 Discussion

6.1 Empirical findings

Regarding usability evaluation, the system preliminarily demonstrated the feasibility and potential of the system for generating usable agent-based VR lecture videos, with high ratings for ease of use, functional integration, and learnability, though consistency and technical support were identified as areas needing improvement.

Meanwhile, the case of the agent-based VR lecture video also received favorable learner responses, as evidenced by positive ratings of learning satisfaction, sense of presence, and agent evaluations, particularly in the HMD condition. Although the embodied agent supported by an integrated plugin was only achieved with a simple visual representation, participants reported acceptable agent evaluation scores. This finding suggests that the current agent with low realism was still considered acceptable within the educational context. This aligns with the uncanny valley perspective, as hyper-realistic characters are prone to falling into the uncanny valley, whereas simply designed humanoid characters may be more favorably received than overly realistic ones (Reinhardt et al., 2020). Nevertheless, future efforts should focus on further refining and enhancing the system's functionalities.

Regarding the main effect of the viewing environment, this study shows that the high-immersion HMD condition significantly enhanced learners' sense of presence, learning satisfaction, and evaluations of the agent compared to the DS condition, but had no significant impact on learning performance or cognitive load. The effect on presence was large in magnitude ($\eta_p^2 = 0.326$), highlighting the technological advantage of HMDs over DS in creating immersive learning environments. This finding is consistent with prior research and aligns with the Cognitive Affective Model of Immersive Learning (CAMIL; Makransky & Petersen, 2021), which posits that learners typically report greater physical presence in VR compared to less immersive media (Allcoat et al., 2021; Makransky et al., 2019a). At the same time, the higher-immersion viewing environment led to higher learning satisfaction, further supporting previous findings on the positive impact of immersion on learning experiences. This may be because the HMDs allow learners to freely choose their viewing perspectives and focal points within the immersive VR environment, thereby enhancing their attentional agency (McGivney, 2025)—the sense of control and autonomy in directing attention. However, this experiential advantage from the high-immersion environment did not translate into superior learning performance compared to the DS, prompting consideration of potential explanations. A possible explanation is related to distraction. Researchers suggest that technologically enhanced spatial presence may increase engagement and hedonic experiences, but these do not necessarily enhance learning (Ahn et al., 2022). In this study, the 360-degree VR video format supports greater immersion through its full FOV. However, this format may also lead to distractions, potentially undermining learning outcomes that require focused attention on specific visual elements within the video (Snelson & Hsu, 2020). Furthermore, the previous study has also shown that the presence of an agent in a VR environment contributes to enhancing user experience but does not significantly improve learning performance (Gao et al., 2024), and it is consistent with the findings of this study. Hence, it is also important to note that the visual presence of an agent may serve as a seductive detail, competing for learners' attention and potentially distracting them from the instructional content (Clark & Choi, 2005). Another possible explanation pertains to the duration of the experiment. Although most studies investigating the educational value of IVR employ short-term interventions of less than 20 min (Klingenberg et al., 2020), the brief intervention (approximately 10 min) used in the present study may have limited the sensitivity of the performance measures. Future research should consider employing longer interventions and multi-

dimensional assessments to better capture nuanced learning gains. Moreover, there was no difference in cognitive load between the HMD and DS groups, which may be attributed to the relatively low difficulty and visual complexity of the instructional materials used in this study. Notably, immersion level influenced learners' perception of the agent. This may be because, in higher-immersive environments, learners are more likely to feel a sense of connection and co-presence with virtual characters. In immersive VR, learners may perceive virtual agents as actual social actors sharing the same virtual space, thereby increasing their sensitivity and responsiveness in evaluating the agent (Won et al., 2023).

Regarding the main effect of prior knowledge, this study shows that learners with higher prior knowledge reported significantly greater senses of presence, learning satisfaction, and evaluations of the agent compared to those with lower prior knowledge, but no significant differences were found in learning performance or cognitive load. Notably, students with higher prior knowledge exhibited greater learning satisfaction, which is consistent with previous findings in the field of multimedia learning (Shangguan et al., 2020). Learners with higher prior knowledge are generally considered to benefit more from multimedia learning environments, as they typically require less external support and possess greater cognitive resources to process new information compared to their lower-knowledge peers (Shangguan et al., 2020). Moreover, prior knowledge not only influences learning performance but also shapes learners' perception, engagement, and attention allocation (Cook, 2006; Dong et al., 2020; Peng et al., 2023). The advantage of high-prior-knowledge learners may allow them to allocate spare attentional capacity to the virtual environment and the agent, leading to enhanced perceptions of presence and more favorable agent evaluations. However, in this study, prior knowledge did not significantly affect learning performance, which contradicts the "knowledge-is-power" hypothesis. The findings may reflect the complex relationship between prior knowledge and learning success—prior knowledge does not always facilitate learning, and it reliably supports new learning only when it is activated, relevant, and coherent (Brod, 2021). Furthermore, factors such as learners' attention, motivation, and interest may moderate the impact of prior knowledge on learning performance (Schneider & Simonsmeier, 2025). The VR instructional material used in this study was relatively low in difficulty, as evidenced by the generally low cognitive load ratings. The material may have failed to engage the interest or attention of learners with high prior knowledge, thereby resulting in non-significant differences in learning performance.

In terms of interaction effects, no significant interaction between the two factors was found across the range of dependent variables examined in this study. This suggests that the advantages of immersive technology are broadly applicable—that is, the impact pattern of immersive technology remains relatively stable across learners with different levels of prior knowledge and is not significantly moderated by learners' initial knowledge base. This indicates that the present study did not find the expertise reversal effect (Kalyuga, 2014), suggesting that the viewing method consistent with immersion principles of multimedia learning (Makransky & Mayer, 2022), which states that immersive virtual environments promote better learning when they incorporate multimedia design principles, is applicable to learners with different levels of prior knowledge.

6.2 Contribution and implication

This study developed an agent-based VR lecture system and filled a gap in the existing literature by examining how viewing environments and prior knowledge influence learning outcomes and experiential factors in immersive video-based learning contexts. From a practical perspective, this study provides a design case that integrates an embodied agent with 360-degree video, offering valuable guidance for educators and instructional designers in creating expressive and engaging immersive educational content. Educational technology developers can focus on the development of VR lecture systems to lower the threshold for the development of VR resources for educational practitioners. Furthermore, this study provides empirical support for the use of 360-degree VR video via HMDs in educational settings to enhance learning experiences. This suggests that educators can adopt such immersive technologies across diverse learner groups with varying levels of prior knowledge, since this study shows that when the design of immersive media follows sound multimedia learning principles, it can transcend differences in “knowledge levels” and become a potential universal and inclusive teaching approach for equity of outcome. Meanwhile, our study did not find a significant effect of viewing environment on learning performance among learners with different levels of prior knowledge. Considering the cybersickness associated with immersive XR headsets, desktop display devices can serve as an accessibility option in educational practice—an inclusive design choice intended for those who are not comfortable using headsets. Moreover, given that prior knowledge did influence learners’ perceived satisfaction in this study, future research should focus on adaptive or scaffolded designs that address individual differences. Such approaches would help ensure that all learners not only achieve comparable outcomes but also experience meaningful and supportive engagement within immersive environments, thereby promoting equity.

6.3 Limitation and future research

Several identified limitations of this study provide avenues for future research. First, participants were recruited from a university for experimental convenience, and the current study has a relatively small sample size that consisted predominantly of female students, which may limit the generalizability of the findings. Future research could expand the sample size to enhance representativeness and further examine longer-term learning retention effects and the moderating effects of other individual difference factors or teaching methods that may influence how learners process information in VR environments (Zhong et al., 2022). Second, although we incorporated several practical strategies to mitigate cybersickness, this experimental study did not include standardized measurements of cybersickness. Future research should employ the simulator sickness questionnaire, such as the cybersickness in virtual reality questionnaire (CSQ-VR; Kourtesis et al., 2023), to formally assess cybersickness and examine its role in VR learning experiences and performance. Third, regarding the design and evaluation of the embodied agent,

this non-interactive agent, which was rendered within the VR video, did not employ intelligent or adaptive gestures synchronized with speech, nor other nonverbal behaviors. Consequently, we relied on simplified instruments from prior research to evaluate the agent. Future design iterations could integrate embodied agents with varying levels of visual complexity and adaptive mechanisms for gesture-speech synchronization to enable further exploration of how to enhance agent expressivity while avoiding the uncanny valley effect. Additionally, future research should consider employing more comprehensive agent evaluation instruments, such as the agent persona instrument (API; Ryu & Baylor, 2005) and the artificial social agent questionnaire (ASAQ; Fitrianie et al., 2025), to holistically assess agent expressivity and provide more robust guidance for the design of embodied agent characteristics. Finally, this study relied on post hoc self-report questionnaires to measure cognitive load in VR learning. Future research could incorporate physiological measures, such as eye-tracking or electroencephalography (EEG), to increase sensitivity to subtle variations in cognitive load. In summary, this study aims to further clarify the interaction between the learning environment type and learners' prior knowledge of VR-based multimedia learning and to offer new insights for both educational practice and future research.

Appendix A

Table 6 Instructional script and contents of VR lecture video

Teaching Segment	Multimedia resource type	Instructional activities and strategies	Brief description of learning contents
1	Slide	Context-based Lesson introduction	Introduction for the start of the VR lecture
2	Slide	Multimedia instruction based on words and pictures on slides	Key concepts and knowledge of five terrain characteristics, elevation and relative height, and contour lines
3	3D animation	3D animation-based multimedia instruction	3D animation presentation for the formation mechanism of the contour lines
4	Slide	Multimedia instruction based on words and pictures on slides	Practical method of drawing contour lines based on surveyed elevation points
5	Slide	Multimedia instruction based on words and pictures on slides	Determine different mountain parts based on contour lines, including Peaks, depressions, valleys, ridges, saddles, cliffs, steep slopes, and gentle slopes
6	Panoramic photo	Virtual field trip based on the 360-degree panoramic environment	Present typical mountain parts in a real scene by a 360° panoramic virtual field trip
7	Slide	Case-based teaching	Calculate the relative height of two places based on the corresponding contour line maps
8	Slide	Summary	Summary of the VR lecture

Appendix B

Table 7 Items of used scales in this study

Scale and label	Item
Learning satisfaction (Cronbach's $\alpha=0.92$)	
LS_1	I was satisfied with this type of VR learning experience.
LS_2	I was satisfied with the learning flexibility and independence of this course.
LS_3	I was satisfied with the instructional activities and strategies in this type of VR learning environment.
LS_4	I was satisfied with this type of VR learning environment.
Presence (Cronbach's $\alpha=0.93$)	
P_1	The virtual environment seemed real to me.
P_2	I had a sense of acting in the virtual environment, rather than operating something from outside.
P_3	My experience in the virtual environment seemed consistent with my experiences in the real world.
P_4	While I was in the virtual environment, I had a sense of "being there".
P_5	I was completely captivated by the virtual scene.
Cognitive load (Cronbach's $\alpha=0.91$)	
CL_1	How difficult do you think it was to learn in this VR lecture video? (mental load)
CL_2	How much effort do you think you devoted to the learning process in this VR lecture video? (mental effort)
Agent evaluation (Cronbach's $\alpha=0.95$)	
AE_1	How friendly do you think the virtual instructor is? (Friendliness)
AE_2	How human do you think the virtual instructor is? (Human-likeness)
AE_3	Did the virtual instructor's face move well while talking? (Face-voice synchronization)
AE_4	Did the virtual instructor's body move well while teaching? (Body movement)
System Usability Scale (Cronbach's $\alpha=0.60$)	
SUS_1	I think that I would like to use this system frequently.
SUS_2	I found the system unnecessarily complex. (R)
SUS_3	I thought the system was easy to use.
SUS_4	I think that I would need the support of a technical person to be able to use this system. (R)
SUS_5	I found the various functions in this system were well integrated.
SUS_6	I thought there was too much inconsistency in this system. (R)
SUS_7	I would imagine that most people would learn to use this system very quickly.
SUS_8	I found the system very cumbersome to use. (R)
SUS_9	I felt very confident using the system. (Satisfaction)
SUS_10	I needed to learn a lot of things before I could get going with this system. (R)
(R) reverse coded	

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Data availability The data that support the findings of this study are available on request from the corresponding author.

Declarations

Competing of interest The authors declare no conflicts of interest.

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